RJA-40 Robotic Joint Actuator Module

Design Document • Rev. B • Ayr Robotics • Doc. No. DD-RJA40-B

1. Overview

The RJA-40 is an integrated brushless-motor joint actuator for collaborative robot arms, combining a BLDC motor, a two-stage planetary/harmonic drivetrain, an absolute encoder, and an onboard joint controller in a single module. It is designed for direct EtherCAT or CAN daisy-chaining across a 6- or 7-axis arm.

2. System Architecture

Figure 1 — RJA-40 Actuator Drivetrain Block Diagram

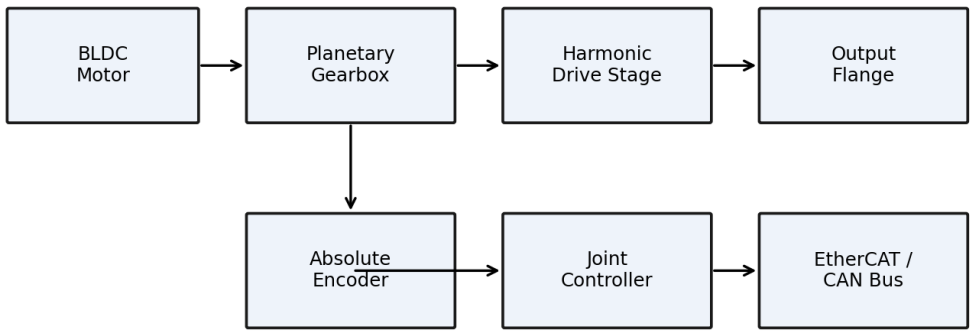


Figure 1 — RJA-40 drivetrain block diagram: motor, gearbox, harmonic drive stage, output flange, encoder feedback, and joint controller.

The absolute encoder reports output-flange position at 16-bit resolution to the joint controller, which closes a local current/velocity/position control loop at 4 kHz and exposes a CiA 402-style profile over EtherCAT or CAN to the arm's central controller.

3. Electrical & Mechanical Specifications

Parameter	Value	Notes
Rated torque	40 N·m	continuous, output flange
Peak torque	120 N·m	1 s max
Gear ratio	101:1	planetary + harmonic stages
Rated output speed	35 rpm	continuous

Operating voltage	48 V DC	±10%
Continuous current	8 A	bus current
Peak current	22 A	1 s max
Encoder resolution	16-bit absolute	output side
Backlash	< 3 arcmin	measured at output flange
Operating temperature	0 °C to 50 °C	housing surface
Weight	1.35 kg	module only
Ingress protection	IP54	shaft seal fitted

Figure 2 — RJA-40 Module (assembly photo)

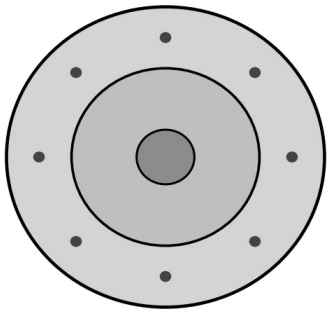


Figure 2 — RJA-40 module, assembled view.

4. Known Limitations

The Rev. B module does not include an integrated torque sensor at the output flange; torque is estimated from motor current, which is less accurate under high-friction or high-backlash conditions. A strain-gauge-based torque sensor is planned for Rev. C.